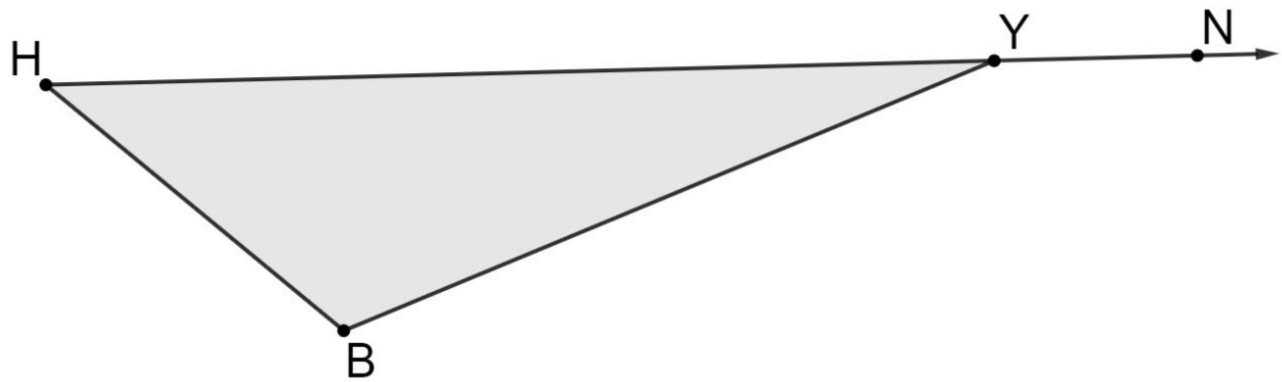
Grade 8	Name____________________
Unit 13 Assessment	Date_________________
Calculator	Period__________

1. Triangle BHY is shown, where $m\angle BHY = (2x + 7)°$, $m\angle HBY = (8x - 18)°$, and $m\angle NYB = (4x + 91)°$.



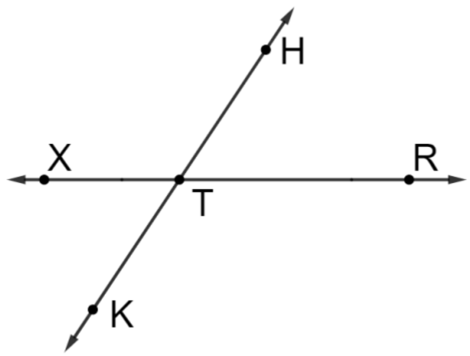
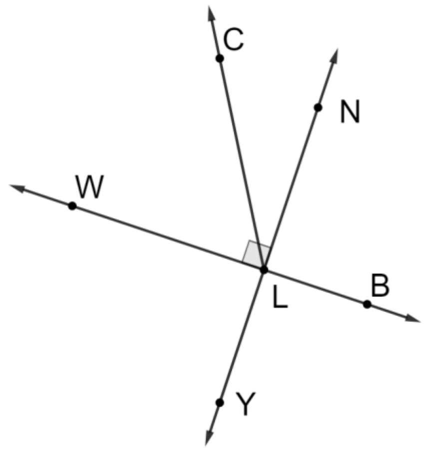
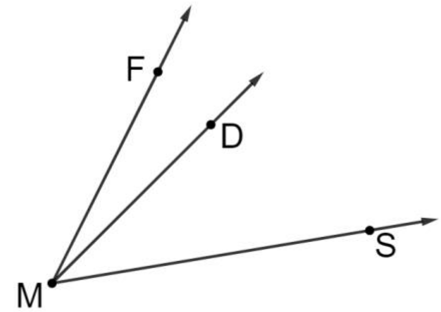
a. Determine the value of x. Show your work.

work
$\boldsymbol{x =}$

b. Determine measure of angle NYB in degrees. Show your work.

work
$\boldsymbol{m\angle NYB}$

2. Three figures are shown.

Figure A	**Figure B**	**Figure C**
		

a. In figure A, name an angle that is congruent to $\angle XTK$.	
b. Name a pair of complementary angles.	
c. In figure C, what type of angle are $\angle DMF$ and $\angle DMS$?	
d. In figure B, name an angle that is supplementary to $\angle NLB$.	

3. The measures of the interior angles of ΔHTR are shown.

$m\angle THR = (4x)°$ $\qquad$ $m\angle RTH = (16x)°$ $\qquad$ $m\angle HRT = (5x)°$

Determine the measure, in degrees, of $\angle HRT$. Show your work.

work	
$\boldsymbol{m\angle HRT}$	

4. Angles MVH and HVK are adjacent complementary angles. The measure of angle HVK is $(6.25x - 10)°$ and the measure of angle MVH is $(2x + 7.6)°$.

a. Determine the value of x. Show your work.

work	
$\boldsymbol{x =}$	

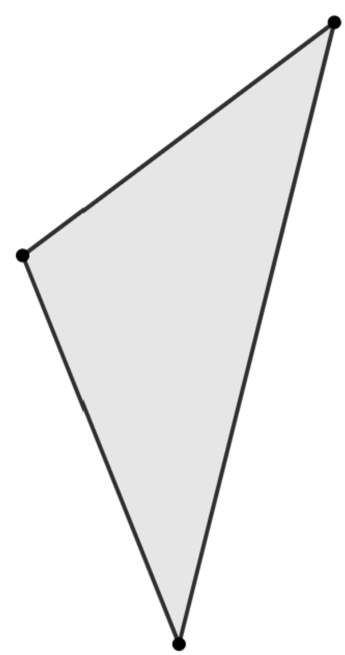
b. Determine measure of angle HVK in degrees. Show your work.

work	
$\boldsymbol{m\angle HVK}$	

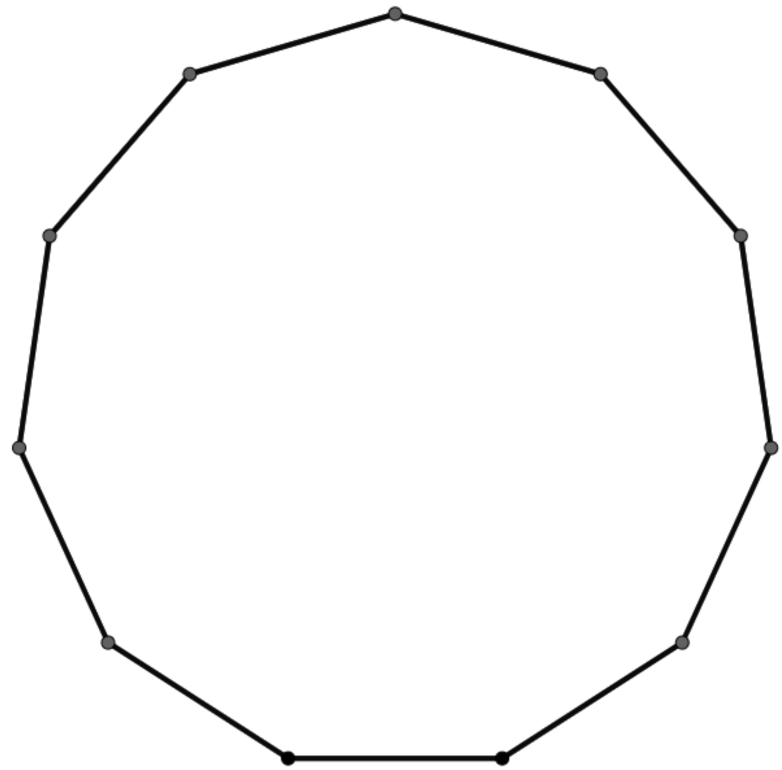
5. A triangle is shown.

a. Draw an exterior angle of the triangle and name it $\angle KRT$.

b. Place an **X** inside the angles whose sum is equal to $m\angle KRT$.



6. A regular polygon is shown.



a. Draw the triangles that can be used to determine the sum of the interior angles of the polygon.

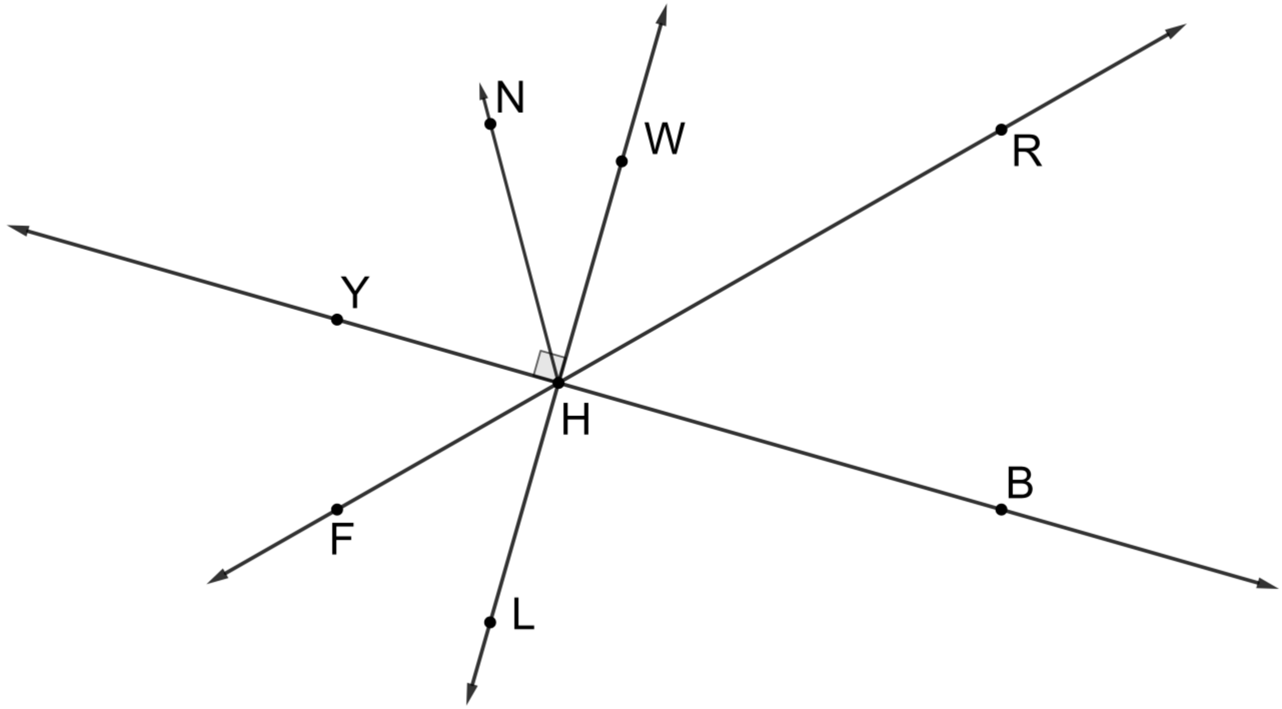
b. Determine the sum of the interior angles in the polygon.

work	
Sum	

c. Determine the measure of one of the interior angles in the polygon.

work	
Measure of One Interior Angle	

7. A figure is shown where $m\angle YHW = 90°$.



a. Name an angle that is supplementary to $\angle BHF$.

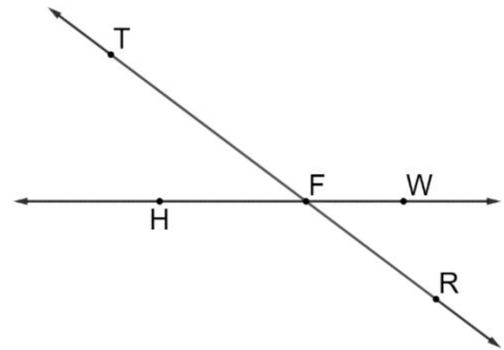
b. Name an angle that is the vertical angle of $\angle YHF$.

c. If $m\angle RHB = 46°$, then find $m\angle FHB$.

8. Determine the sum, in degrees, of the interior angles in a regular polygon with 21 sides.

work	
Sum of Interior Angles	

9. Lines TR and HW are shown, where $m\angle HFT = (5x - 7.5)°$ and $m\angle HFR = (15x + 13.5)°$.



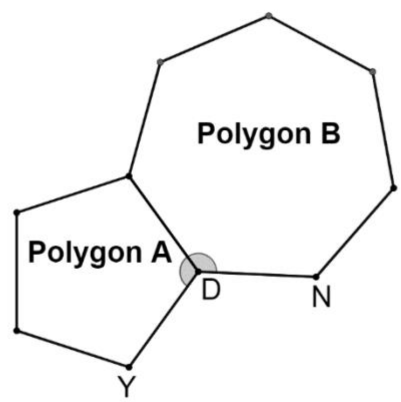
Determine $m\angle WFR$, in degrees. Show your work.

work	
$m\angle WFR$	

10. Two polygons are shown, where polygon A has 5 sides and polygon B has 7 sides. Angle YDN is marked, as shown.

Determine the measure, in degrees, of angle YDN. Round to the nearest thousandth.

Show your work.



work	
$m\angle YDN$	